

AN ELEMENTARY PROOF OF ERLBACHER'S COUNTEREXAMPLE TO TXGRAFFITI CONJECTURE 9

ABSTRACT. Erlbacher refuted Conjecture 9 of Davila, generated by TxGraffiti, with a connected cubic triangle-free graph G on 14 vertices satisfying $\gamma(G) = 4$ and $Z(G) = 7$, and established in the same artifact that the excess $Z - \gamma$ reaches 5 in a chain family inside the same class, so the proposed bound $Z(G) \leq \gamma(G) + 2$ for connected cubic diamond-free graphs fails neither by one unit nor at one graph. This note contributes no new example and no new bound: for Erlbacher's 14-vertex graph we prove the two lower bounds $\gamma(G) \geq 4$ and $Z(G) \geq 7$ without computation, the second from an explicit family of forts, forts being the route Erlbacher identifies, in place of the exhaustive subset searches of the artifact.

1. THE COUNTEREXAMPLE

All graphs are finite, simple, and undirected. A set $D \subseteq V(G)$ is *dominating* if $N[D] = V(G)$, and the minimum size of such a set is the *domination number* $\gamma(G)$. Starting from a set of blue vertices, the zero forcing rule allows a blue vertex with exactly one white neighbor to force that neighbor blue. The minimum size of an initial set that can color all vertices blue is the *zero forcing number* $Z(G)$; see [1].

Davila posed the following as Conjecture 9 of [2], generated by TxGraffiti:

$$(1) \quad Z(G) \leq \gamma(G) + 2$$

for every connected cubic diamond-free graph G . Erlbacher [3] refuted it. The cited artifact exhibits the graph below, establishes $\gamma(G) = 4$ and $Z(G) = 7$ by exhaustive search over vertex subsets in two independent implementations, with mutation tests and an independent re-verification, and records explicit witnesses for both upper bounds. It also fixes the statement against its source and settles the ambiguity in the hypothesis: “diamond” is defined there as a subgraph while F -freeness is defined by induced subgraphs, and a triangle-free graph is diamond-free under either reading.

The artifact does not stop at one graph. For the chain family obtained from k copies of $K_{3,3}$ by subdividing one edge of each end block and two edges of each middle block and joining consecutive subdivision vertices by bridges—connected, cubic and triangle-free, on $8k - 2$ vertices—it determines γ and Z exactly for $2 \leq k \leq 5$ and γ exactly for $k \leq 8$. The excess $Z - \gamma$ is thus 4 at $n = 30$ and 5 at $n = 38$ inside the same class, so no bound

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$Z \leq \gamma + c$ with $c \leq 4$ survives either, and the failure of (1) is not confined to one exceptional graph. Erlbacher records the unbounded-excess statement as a conjecture only, and identifies forts as the tool for lower bounds on Z in this family.

The present note adds no counterexample and no new bound. The graph, both values, the dominating set used below, and forts as the route to the lower bound on Z are all Erlbacher's. What is new here is computation-free proofs of the two lower bounds $\gamma(G) \geq 4$ and $Z(G) \geq 7$ for the 14-vertex graph, which replace the exhaustive subset searches of the artifact for that one graph.

Let

$$\begin{aligned} A &= \{a_1, a_2, a_3\}, & B &= \{b_1, b_2, b_3\}, \\ C &= \{c_1, c_2, c_3\}, & D &= \{d_1, d_2, d_3\}. \end{aligned}$$

Define G on $A \cup B \cup C \cup D \cup \{x, y\}$ by

$$\begin{aligned} E(G) &= \{a_i b_j : 1 \leq i, j \leq 3, (i, j) \neq (3, 1)\} \\ &\cup \{c_i d_j : 1 \leq i, j \leq 3, (i, j) \neq (3, 1)\} \\ &\cup \{a_3 x, x b_1, c_3 y, y d_1, xy\}. \end{aligned}$$

Equivalently, take two copies of $K_{3,3}$, subdivide one edge in each, and join the two subdivision vertices.

Theorem 1. *The graph G is connected, cubic, and triangle-free, and*

$$\gamma(G) = 4 \quad \text{and} \quad Z(G) = 7.$$

Consequently, G is diamond-free and refutes (1).

A nonempty set $F \subseteq V(G)$ is a *fort* if every vertex outside F has either zero or at least two neighbors in F .

Lemma 2. *Every zero forcing set intersects every fort.*

Proof. If a zero forcing set S were disjoint from a fort F , then immediately before the first force into F , the forcing vertex would lie outside F and have exactly one neighbor in F , contrary to the definition of a fort. $\square$

Proof of Theorem 1. The construction makes G connected and cubic. Each subdivided copy of $K_{3,3}$ is triangle-free, and the edge xy is a bridge, so G is triangle-free. Since every diamond contains a triangle, G is diamond-free.

The set $\{a_3, b_1, c_3, d_1\}$, the dominating witness recorded in [3], dominates G , so $\gamma(G) \leq 4$. Since G has 14 vertices and maximum degree 3, a set of k vertices dominates at most $4k$ vertices. Hence $\gamma(G) \geq \lceil 14/4 \rceil = 4$, and therefore $\gamma(G) = 4$.

For the zero forcing upper bound, let

$$S_0 = \{y, a_1, a_2, b_2, c_1, c_2, d_2\}.$$

The sequence

$$b_2 \rightarrow a_3, \quad d_2 \rightarrow c_3, \quad c_3 \rightarrow d_3, \quad c_1 \rightarrow d_1, \quad y \rightarrow x, \quad a_3 \rightarrow b_3, \quad a_1 \rightarrow b_1$$

is valid, so $Z(G) \leq 7$.

It remains to prove $Z(G) \geq 7$. Put $U = A \cup B$ and $W = C \cup D$. The following six subsets of U are forts:

$$\{a_1, a_2\}, \quad \{b_2, b_3\},$$

and

$$\{a_i, a_3, b_1, b_j\}, \quad i \in \{1, 2\}, \quad j \in \{2, 3\}.$$

For either two-element set, each outside vertex adjacent to it has exactly two neighbors in it. For a four-element set, the only outside vertices with neighbors in the set are x , the other vertex of $\{a_1, a_2\}$, and the other vertex of $\{b_2, b_3\}$; each has exactly two such neighbors.

No two vertices meet all six forts. Indeed, any two-vertex set meeting the first two forts must be $\{a_i, b_j\}$ with $i \in \{1, 2\}$ and $j \in \{2, 3\}$, but it misses the four-element fort obtained from the other choices of i and j . Lemma 2 therefore gives

$$|S \cap U| \geq 3$$

for every zero forcing set S . By symmetry, $|S \cap W| \geq 3$.

Suppose that S is a zero forcing set of size six. Then

$$|S \cap U| = |S \cap W| = 3 \quad \text{and} \quad S \cap \{x, y\} = \emptyset.$$

Define

$$\mathcal{P} = \{\{a_1, a_3\}, \{a_2, a_3\}, \{b_1, b_2\}, \{b_1, b_3\}\},$$

$$\mathcal{Q} = \{\{c_1, c_3\}, \{c_2, c_3\}, \{d_1, d_2\}, \{d_1, d_3\}\}.$$

For every $p \in \mathcal{P}$ and $q \in \mathcal{Q}$, the set

$$\{x, y\} \cup p \cup q$$

is a fort. If $p = \{a_i, a_3\}$, the other vertex of $\{a_1, a_2\}$ has zero neighbors in $p \cup \{x\}$, while b_1, b_2, b_3 each have two. If $p = \{b_1, b_j\}$, the other vertex of $\{b_2, b_3\}$ has zero, while a_1, a_2, a_3 each have two. The verification for q is symmetric.

Since S misses x and y , Lemma 2 implies that S meets p or q for every $p \in \mathcal{P}$ and $q \in \mathcal{Q}$. Hence S meets every member of $\mathcal{P}$, or it meets every member of $\mathcal{Q}$. In the first case, together with the two two-element forts above, $S \cap U$ meets all six pairs

$$\{a_1, a_2\}, \{a_1, a_3\}, \{a_2, a_3\}, \quad \{b_1, b_2\}, \{b_1, b_3\}, \{b_2, b_3\}.$$

These are the edges of two vertex-disjoint copies of K_3 , so at least two vertices from each copy are required. Thus $|S \cap U| \geq 4$, a contradiction. The second case similarly gives $|S \cap W| \geq 4$. Therefore no six-vertex set is zero forcing, so $Z(G) \geq 7$. Together with the forcing sequence, this proves $Z(G) = 7$. $\square$

The edge xy is a bridge, but bridgelessness is not a hypothesis of Conjecture 9. Moreover, x and y are centers of induced claws, so the example does not affect Davila's theorem for connected cubic claw-free graphs.

REFERENCES

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